

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Build multi-agent systems (LangGraph, CrewAI, AutoGen, OpenAI Agents SDK) with tool/function calling and structured JSON outputs, deployed to production at 99%+ uptime.
- Combine GraphRAG with vector-based RAG (Pinecone) for grounded semantic search and LLM integration on complex enterprise queries.
- Built a deployment-gated evaluation framework enforcing guardrails and observability across LLM/agent outputs, blocking 15% of failed releases before production.
- Build platform APIs (Python, TypeScript) partnering with data science and software engineering teams end-to-end.

QureAI (AI Startup)

Mar 2026 – Present

ML Infrastructure Engineer

- Developed deep learning models (PyTorch) with domain-specific neural network architectures, optimizing performance for production inference.
- Built GraphRAG pipelines over Neo4j knowledge graphs combined with vector retrieval, improving semantic search recall 61% to 89% on a 2K-query eval set.
- Built custom evaluators for RAG/agent workflows tracking drift, latency, and regression across model versions, catching failures before they reached production.

Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

Software Engineer

- Architected and operated 25+ production C++ and SQL-backed microservices for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Standardized failure-handling contracts (retry, timeout, propagation semantics) across 25+ services, reducing production escalations 60% in one quarter.
- Mentored engineers through code reviews and design write-ups, driving consistent engineering standards.

Projects

LangFetch | tinyurl.com/LangFetch

- LangGraph agent with MCP-compatible tool-use compiling natural-language questions into dependency-ordered SQL subqueries with Pinecone vector retrieval and hybrid re-ranking.

RootCause AI | github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports.

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Production LLM eval platform with hallucination detection, drift tracking, rubric-based scoring, and regression alerting across model versions; FastAPI backend, PostgreSQL.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, TypeScript, SQL, Java, C++

Agentic AI & LLMs: LangGraph, CrewAI, AutoGen, OpenAI Agents SDK, Tool/Function Calling, Structured Outputs

RAG & Knowledge Graphs: RAG, GraphRAG, Neo4j, Knowledge Graphs, Pinecone, Semantic Search, PyTorch

Evaluation & Cloud: Model/Agent Evaluation, Guardrails, Observability, AWS, GCP, Docker, CI/CD

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer